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Invited review: Unlocking growth and development potential in dairy calves through precision protein feeding

M. H. Ghaffari,[1]* J. K. Drackley,[2]* and A. F. Kertz[3]
[1]Institute of Animal Science, University of Bonn, 53115 Bonn, Germany
[2]Department of Animal Sciences, University of Illinois, Urbana, IL 61801
[3]ANDHIL LLC, St. Louis, MO 63122

ABSTRACT

Optimizing protein nutrition is pivotal for enhancing growth, metabolic efficiency, and long-term productivity of dairy calves. This review summarizes current research on precision protein feeding based on the NASEM (2021) model, which aligns MP and ME with calf developmental stages to optimize nutrient utilization. In the preweaning phase, the CP requirement of Holstein calves (<125 kg BW) increases with increasing ADG, with a 50-kg calf requiring 102 g of CP (18.2% DMI) for 0.2 kg/d ADG and 315 g of CP (25.5% DMI) for 1.0 kg/d ADG, with ME intake increasing from 2.56 to 5.66 Mcal/d. The review underscores the importance of adjusting the CP-to-ME ratio; adequate supply of limiting AA such as Lys, Met, and Thr; and a balanced ratio of RDP and RUP to optimize growth and metabolism. Amino acids are essential for muscle development, immune response, and enzyme function. Protein intake in the preweaning phase improves growth performance, weight gain, and feed efficiency, although excess protein in the diet can reduce overall growth efficiency. The balance between RDP and RUP is critical for microbial protein synthesis and nitrogen retention, with RDP becoming the dominant protein source after weaning due to increased reliance on microbial protein. Furthermore, macronutrient-specific adjustments in milk replacers, such as the addition of protein, fat, or lactose, lead to different metabolic responses, with protein primarily supporting lean tissue building, whereas fat and lactose promote energy storage. These findings emphasize the need for precise nutritional strategies that consider protein quality and AA availability to ensure optimal growth and metabolic efficiency. Furthermore, protein levels in calf starter diets, ideally between 20% and 23% CP (DM basis), lead to optimal growth, especially when combined with moderate to high milk feeding, although higher protein levels are not always growth promoting, highlighting the need for precise nutritional adjustments according to the developmental needs of the calf. For growing heifers, protein nutrition after weaning continues to be a critical factor in optimizing growth, efficiency, and later lactation performance. Stage-specific adjustments to the balance between RDP and RUP are important to support microbial protein synthesis and nitrogen retention, with RDP becoming the dominant protein source as heifers mature and transition to a more energy-driven fat deposition phase. These findings are critical for refining protein and energy utilization strategies, optimizing calf development, and supporting long-term lactation performance. In summary, this review provides insights into the optimal protein feeding strategies to improve calf growth and metabolic efficiency in the pre- and postweaning phases.
Key words: protein requirements, metabolizable protein and energy, growth efficiency, calf starter

INTRODUCTION

Optimizing early-life nutrition is crucial for the long-term health, metabolic efficiency, and productivity of dairy calves (Van Amburgh et al., 2014). During the preweaning phase, protein and energy intake should be balanced to support lean tissue growth while preventing excess fat deposition (Gerrits et al., 1996; Amado et al., 2024b). Advances in calf nutrition emphasize the importance of precisely formulated milk replacers (**MR**) and starter feeds that optimize both protein and energy supply. Beyond total CP intake, factors such as age, BW, protein digestibility, AA profile, and the balance between RDP and RUP significantly influence protein utilization efficiency in dairy calves (van den Borne et al., 2006a; Oney et al., 2016; Gerrits, 2019). Research has shown that variations in CP and ME supply directly influence BW gain, feed efficiency, and N retention (Lammers and Heinrichs, 2000; NASEM, 2021; Stamey Lanier et

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*Corresponding authors: morteza1@uni-bonn.de and drackley@illinois.edu

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al., 2021). This review summarizes current knowledge on the protein and energy requirements of dairy calves, focusing on the dynamics of MP, the AA balance, and the interaction between CP, ME, and protein fractions (RDP and RUP) at different stages of development. It also highlights the influence of starter feed composition and processing on protein utilization and identifies key research gaps in optimizing N efficiency and lean growth during the pre- and postweaning periods.

PROTEIN AND ENERGY REQUIREMENTS FOR GROWTH EFFICIENCY IN CALVES

Proper nutrition in the preweaning phase is crucial for optimal growth and development of calves. Calves, defined as animals weighing less than 18% of their mature BW (for Holsteins, approximately under 125 kg, with a mature BW of 700 kg), require a carefully balanced diet to meet their specific energy and protein requirements. In addition, the use of empty BW (**EBW**) instead of live BW for nutrient calculations in NASEM (2021) provides more accurate metabolic scaling that aligns nutrient intake with metabolic tissue and not just overall body mass. The NASEM model includes conversion factors for EBW: 0.94 for milk-fed calves, 0.93 for mixed-fed calves, and 0.85 for weaned calves, which helps to refine nutrient requirements at different feeding stages.

Refining energy calculations is equally important to meet the metabolic needs of growing calves. NASEM (2021) estimates net energy for maintenance at 0.097 Mcal/kg $EBW^{0.75}$ and ME for maintenance at 0.138 Mcal/kg $EBW^{0.75}$, both of which are higher than NRC (2001) estimates but consistent with recent experimental data. These values consider the increasing metabolic requirements as calves grow and the shifts in their nutrient requirements over time. In particular, the inclusion of scurf protein losses and a reduced efficiency factor of 0.68 for scurf and metabolic fecal protein in the NASEM model further refines maintenance protein requirements. The NASEM (2021) guidelines also highlight the critical relationship between ADG, DMI, and protein requirements in preweaning calves. For example (Table 1), a 50-kg Holstein calf with an ADG of 0.2 kg/d requires 102 g of CP daily (18.3% of DMI), whereas a calf with an ADG of 1.0 kg/d requires 315 g of CP per day (25.6% of DMI). This shows how the protein requirement increases with an increasing growth rate. In addition to the protein requirement, the ME requirement also increases from 2.56 to 5.66 Mcal/d, underlining the importance of balanced protein and energy intake for optimal growth. For calves with lower ADG targets (0.2–0.4 kg/d), MR formulations with ~20% CP may be sufficient. However, for calves with higher ADG targets (≥0.8 kg/d), MR formulations with 26% CP (DM basis) or more are necessary to meet their increased protein needs. For instance, a high-protein MR formulation described by Wilms et al. (2022) contains 26% protein, 18% fat, and 41.5% lactose, specifically designed to support the growth of calves with elevated ADG targets.

Table 1. Energy and protein requirements for a 50-kg Holstein calf fed milk replacer (thermoneutral conditions) based on NASEM (2021) equations[1]

ADG (kg/d)	DMI (kg/d)	ME (Mcal/d)	CP (g/d)	CP (% of DM)
0.20	0.56	2.56	102	18.2
0.40	0.72	3.29	156	21.6
0.60	0.89	4.05	209	23.7
0.80	1.06	4.85	262	24.7
1.00	1.24	5.66	315	25.5

[1]Values are based on a milk replacer containing 4.58 Mcal ME/kg DM.

In a recent study by Bartlett et al. (2024), the effects of different MR intakes (1.25%, 1.75%, and 2.25% of BW) on growth, body composition, and EBW composition in young dairy calves were investigated without feeding a starter feed. Calves fed greater MR intakes had greater final weight (58.6 kg, 70.6 kg, and 82.9 kg), ADG (0.36 kg, 0.70 kg, and 1.03 kg/d), and final heart girth, body length, and gain-to-feed ratio (Bartlett et al., 2024). Calves fed greater levels of MR had greater lean tissue and fat mass, with a linear decrease in water (from 72.2% to 67.8%) and protein (from 17.6% to 14.5%) and an increase in fat (from 10.2% to 17.7%; Bartlett et al., 2024). The efficiency of energy use (retained energy: gross energy intake) improved linearly with greater MR intake, whereas retained energy:ME for growth remained unchanged (Bartlett et al., 2024). Jaeger et al. (2020) demonstrated that Holstein heifer calves fed a texturized calf starter (40% pellets and 60% grains with 18% CP, DM basis) together with different MR treatments (20% or 24% CP, DM basis, at feeding rates of 0.57 or 0.68 kg/d) had greater ADG (e.g., 0.96–1.06 kg/d during d 43–56) and better feed conversion ratio both before and after weaning when fed at higher MR rates (0.68 kg/d).

The relationship between CP and ME intake in dairy calves shows a complex interaction that influences growth performance. Hill et al. (2008a) found that calves fed a 20% CP (as-fed basis) MR at 440 g DM/d or a >25% CP MR at >660 g DM/d achieved optimal ADG on textured starter diets containing 20.5% CP (63 g CP/Mcal ME). For calves 2 to 4 mo of age, 17% CP or 52 g CP/Mcal ME maximized ADG, whereas 18% CP or 56 g CP/Mcal ME optimized feed efficiency (**FE**). Similarly, Hill et al. (2009) showed that the optimal CP-to-ME ratio to maximize ADG was 51.5 g CP/Mcal ME at the low MR intake (25% CP, 17% fat) and 55.0 g CP/Mcal ME at the high MR intake (27% CP, 17% fat; DM basis), where the starter consisted of 37% rolled corn, 35% protein pellets, 25% whole oats, and 3% molasses (Hill et al., 2009).

These results are in line with those of Rauba et al. (2019), who investigated the effects of protein and energy intake by MR and starter on the growth and first-lactation performance of Holstein heifer calves. In the study by Rauba et al. (2019), the calves were fed a texturized feed with CP content of 18% and ME of 3.28 Mcal/kg (DM basis) as starter feed. The feed was offered ad libitum from 6 wk of age, with intakes varying from 2.72 to 3.2 kg/d, and then switched to a cereal mixture with a CP content of 16% (as fed) at 2 mo of age, together with hay ad libitum, mainly alfalfa or a mixture of alfalfa and grass. A total of 4,534 calves were fed 20% CP, 20% fat MR at an intake of 0.57 kg/d or more, with 85% receiving the standard diet (Rauba et al., 2019). The study found that calves with ADG >0.80 kg/d consumed more protein and ME, with 6-wk intake averages of 4.7 ± 1.0 kg for MR protein and 3.6 ± 1.5 kg for starter protein (Rauba et al., 2019). Calves with greater ADG (>0.80 kg/d) consumed more starter feed, starter protein (9.5 ± 2.6 kg at 8 wk), and ME (206.6 Mcal), leading to increased 305-d milk, fat, and protein production (Rauba et al., 2019). The optimal CP-to-ME ratio for maximum ADG was 51.5 g CP/Mcal of ME for lower MR intake and 55.0 g CP/Mcal for greater MR intake (Rauba et al., 2019).

DYNAMIC SHIFTS IN MACRONUTRIENT OXIDATION AND NITROGEN RETENTION DURING CALF DEVELOPMENT

The efficiency of protein utilization in growing dairy calves is determined by the relative contributions of protein, fat, and carbohydrate oxidation to ME and by developmental shifts in tissue accretion patterns (Gerrits, 2019; Van Amburgh et al., 2019). In the preweaning phase, calves primarily oxidize lactose and fat for energy, whereas the oxidation of AA remains minimal due to the high biological value of milk proteins and the priority of lean tissue growth (Gerrits, 2019; Amado et al., 2024a). As a result, excess dietary protein is often metabolized during this phase, contributing to increased urea synthesis and reduced N efficiency when AA supply exceeds tissue requirements (Amado et al., 2024b).

In the early postweaning phase, a significant proportion of stored energy supports protein deposition, particularly in the gastrointestinal tract, which accounts for up to 24% of retained protein compared with 14% to 20% at more mature stages, while fat deposition remains low (Meyer, 2005; Stamey et al., 2012; Van Amburgh et al., 2019). Due to the high energetic cost of protein breakdown, resulting from protein turnover, AA deamination, and urea synthesis, the efficiency of ME utilization for protein gain is much lower than that of fat or carbohydrate (Gerrits, 2019). As heifers progress toward puberty (150–350 kg BW), retained energy rises significantly to 3.1 Mcal/kg EBW, driven by increased fat deposition (from 110 to 230 g/kg empty body gain, **EBG**), while protein retention remains relatively constant at 180 g/kg EBG up to 250 kg BW, decreasing by ~30 g/d near puberty as fat displaces protein in the gain (Van Amburgh et al., 2019).

Increasing protein content in MR has also been shown to directly influence growth efficiency. Blome et al. (2003) showed that increasing the protein concentration in MR from 16.1% to 25.8% CP (DM basis) resulted in a linear improvement in ADG (mean range: 0.38–0.62 kg/d) and gain-to-feed ratio (mean range: 0.51–0.78), demonstrating that higher dietary CP content improves growth efficiency and feed utilization in calves without access to hay or starter. Increased N retention and absorbed N were also observed with higher dietary protein, suggesting improved N utilization for growth and lean tissue deposition, while fat content decreased (Blome et al., 2003).

Feeding strategies that enhance dietary protein supply can improve growth performance and N efficiency, particularly during early development. In milk-fed calves supplemented with low-protein solid feed (**SF**; 93 g CP/kg DM at levels of 0, 9, 18, or 27 g DM/kg $BW^{0.75}$ per day), N retention increased linearly by 0.55 g N/g DM solid feed, with 19% of this improvement (0.10 g of N) directly attributable to urea recycling (Berends et al., 2014). This enhanced retention was accompanied by a 43% reduction in urinary urea N excretion and increased gastrointestinal entry rate of urea, paralleled by upregulation of ruminal urea transporter B, indicating greater capacity for urea transport into the rumen (Berends et al., 2014). Furthermore, in veal calves (23 wk of age, 180 ± 3.7 kg BW), fed equal total N intake (1.8 g N/kg $BW^{0.75}$ per day), increasing the proportion of dietary N from low-protein solid feed (93 g CP/kg DM) from 12% to 22% shifted N excretion from urine to feces, increased urea recycling, and reduced total-tract NDF digestibility by 9%, but did not improve protein retention (0.71 g N/kg $BW^{0.75}$ per day) or anabolic use of recycled urea (Berends et al., 2015). However, when solid feed provided 36% of total N intake and was higher in protein, protein retention increased by 17%, urea production and excretion declined, and NDF digestibility improved by over 10%, indicating that higher RDP in SF enhances microbial growth and N utilization efficiency (Berends et al., 2015). These findings underscore the importance of not only protein quantity but also the route, timing, and fermentability of dietary N for efficient utilization.

Macronutrient-specific adjustments in MR reveal distinct metabolic responses and utilization efficiencies in 21-d-old Holstein-Friesian calves, as explored by Amado et al. (2024a,b). Calves fed isoenergetic MR treatments, control (23.3% CP, 21.2% fat, and 48.8% lactose, DM basis; 550 kJ/kg $BW^{0.85}$ per day) or control plus 125 kJ/kg $BW^{0.85}$ as protein (+PRO), fat (+FAT), or lactose (+LAC). They found that +PRO calves achieved the highest BW gain (567 g/d) and N retention (818 mg N/kg $BW^{0.85}$), retaining 52% of added protein as body protein, whereas +FAT and +LAC calves stored 67% and 49% of extra energy as fat, with minimal protein accretion (5% and 0%, respectively; Amado et al., 2024b). Gross energy efficiency (retained energy/gross energy intake) peaked in +FAT (34%), followed by +LAC (30%) and +PRO (29%), whereas digestible N efficiency was highest in +LAC (89%) and lowest in +PRO (68%), despite their greater N retention, suggesting that excess protein is catabolized rather than stored (Amado et al., 2024b). Metabolically, +PRO calves exhibited elevated heat production (424 kJ/kg $BW^{0.85}$ per day) and fat oxidation (6.3 kJ/d), whereas +LAC calves showed a higher respiratory quotient (0.95) and carbohydrate oxidation (14.3 kJ/d), reducing fat oxidation to 2.4 kJ/d; fasting heat production (~312 kJ/kg $BW^{0.85}$) and glucose oxidation ($^{13}CO_2$ recovery: 75%–80%) remained stable across groups, indicating that macronutrient effects are postprandial metabolism rather than basal energy expenditure (Amado et al., 2024a). These findings demonstrate that protein drives lean growth, whereas fat and lactose favor energy storage, guiding precision nutrition strategies for early calf development (Amado et al., 2024a,b). Notably, the incremental efficiency of protein utilization declines with age, decreasing from ~70% at birth to 55% at 200 kg BW (NASEM, 2021), and ranging from 20% to 40% in veal calves above 150 kg (Gerrits et al., 1996; van den Borne et al., 2006b; Labussière et al., 2008). These dynamics underscore the need for age-specific feeding programs.

ROLE OF AA IN CALF NUTRITION

Amino acids are indispensable for dairy calves, serving as substrates for protein biosynthesis critical to muscle development, enzyme synthesis, and immune function (Kimball and Jefferson, 2002; Wu et al., 2014; Cao et al., 2019). Unlike mature ruminants, for which microbial protein contributes substantially to AA supply, preweaning calves rely on dietary protein from milk or MR due to their nonfunctional rumen (NASEM, 2021). Commercial MR formulations typically target CP levels of 20% to 28% of DM, but often optimize AA profiles less, resulting in inefficient N utilization where excess AA are oxidized, increasing plasma urea N and energy expenditure (Gerrits et al., 1996; Amado et al., 2024b). Milk proteins, such as whey and casein, are the benchmark for MR, offering ~94% apparent ileal digestibility and a balanced EAA profile (Quigley et al., 1985; Thornsberry et al., 2016). Hill et al. (2008b) demonstrated that milk proteins alone may not meet Lys and Met demands for calves targeting an ADG >0.5 kg/d, and supplementation to 2.34% Lys (17 g/d) and 0.72% Met in a 26% CP (as-fed basis) whey-based MR (0.68 kg/d) increased ADG by 17% and 13%, respectively. However, Castro et al. (2016) observed that additional Met might not be necessary when MR contains more than 28% CP with 0.57% Met. Moreover, low BW gain due to stress or illness can blunt the positive effects of precise AA formulation (da Silva et al., 2018).

Multiple studies have explored optimal AA concentrations for preweaning calves (Williams and Hewitt, 1979; Tzeng and Davis, 1980; Klemesrud et al., 2000; Hill et al., 2007b, 2008b; Wang et al., 2012; Bittar et al., 2018; van Keulen et al., 2020; Silva et al., 2021). Lysine, Met, and Thr consistently emerge as limiting EAA in MR. Requirements differ with calf age, growth rate, and health (Williams and Hewitt, 1979; Hill et al., 2007b, 2008b; Zubia et al., 2023). Williams and Hewitt (1979) estimated Lys needs at 7.8 g/d for dairy calves, based on N retention, plasma AA, and 0.25 kg/d growth. Carcass analysis defined EAA needs (g/d): Met 2.1, Cys 1.6, Thr 4.9, Val 4.8, Ile 3.4, Leu 8.4, Tyr 3.0, Phe 4.4, His 3.0, Arg 8.5, and Trp 1.0, noting an Arg shortage in cow milk (Williams, 1978). van Weerden and Huisman (1985) identified Met + Cys and Lys as the first limiting AA in veal calves (55–70 kg, 5–7 wk, 875–1,000 g/d gain) fed skim milk powder-based MR, with Thr and Ile as the next most limiting AA. Maximum N deposition needed 9.2 g/d Met + Cys AA (Met:Cys 2.9:1, 0.74%, 21.0% CP), 23 g/d Lys (1.81%, 22.0% CP), Thr (0.9%), and Ile (1.1%) at 20.0% CP, with Leu adequate at ~18.0% CP in the form of skim milk powder (van Weerden and Huisman, 1985). Hill et al. (2008b) found Lys and Met in milk proteins suboptimal for Holstein calves (<5 wk, 48 kg BW) fed whey-based MR (24%–28% CP, 17% fat, as-fed basis; 0.68 kg/d). Supplementation to 2.34% Lys (17 g/d) raised ADG by 17%, and 0.72% Met by 13% (linear) and 7% (quadratic). However, Thr supplementation beyond a 0.60 Thr:Lys ratio conferred no additional benefits (Hill et al. 2008b). These results suggested an optimal Met:Lys ratio of 0.31 and a Met+Cys:Lys ratio of 0.54 at a dietary intake of 204 g CP/d (Hill et al. 2008b). More recently, Ahangarani et al. (2020) supplemented a MR (25.1% CP, 20.3% fat) with 0.3% glutamic acid while calves had access to pelleted starter feed (19.3% CP, DM basis), resulting in minor changes in AA metabolism, including increased plasma Leu and slight adjustments in Ile and Met. However, this supplementation did not improve calf performance or affect gut permeability (Ahangarani et al., 2020). This study, consistent with that of Yeste et al. (2020), found that supplementing MR with 4.5 g/d of Trp for Holstein calves (48–62 d old) increased plasma Trp and kynurenine levels but did not improve growth, intake, or behavior during weaning while calves had free access to chopped barley straw and pelleted starter feed (19.3% CP, DM basis).

Whey and milk proteins serve as primary protein sources in calf MR formulations due to their high quality, but increasing human demand for milk proteins has intensified competition and increased costs, leading to the exploration of plant-based alternatives such as soy and wheat (Thornsberry et al., 2016; Ansia and Drackley, 2020). Despite improvements in processing, these alternatives often have unbalanced AA profiles and antinutritional factors (Thornsberry et al., 2016; Ansia and Drackley, 2020). Plant-based proteins such as soy and wheat pose a challenge for MR formulation due to their influence on digestibility and AA profile, despite advances in processing. Soy protein increases indigestible endogenous protein synthesis (Montagne et al., 2000) and contains antinutritional factors (e.g., protease inhibitors) that impair intestinal digestion and absorption (Lallès et al., 1999; Ansia and Drackley, 2020). Variations in nonantigenic protein traits, such as AA concentration, peptide size, and structure, influence differences in digestibility and performance between milk- and soy-based MR (Ansia and Drackley, 2020). However, depending on the soy product and inclusion rate, soy-based MR may exhibit reduced disparities or higher levels of specific AA, including Cys, Arg, Gly, Asp, Ala, and Phe (Ansia and Drackley, 2020).

Kanjanapruthipong (1998) found that replacing milk protein with soy protein in MR can reduce BW gain in calves. However, supplementing an MR with 43% soy protein with Lys, Met, and Thr significantly increased ADG by 62.4 g/d and improved N retention compared with an MR with skim milk protein (Kanjanapruthipong, 1998). Chagas et al. (2018) found that 8.0 g/d of Met plus Cys was sufficient for calves from 8 to 30 d, whereas 8.41–9.81 g/d increased performance and protein gain in calves from 30 to 60 d, with no effect on nutrient digestibility across 4 Met + Cys inclusion levels (8.0, 8.7, 9.4, and 10.2 g/d) when no starter was fed. Wheat protein, with 2.5% Lys and 40% less Thr in hydrolyzed wheat gluten than skim milk (Thornsberry et al., 2016), achieved performance equivalent to milk-based MR when formulated as 4.5% hydrolyzed wheat protein with Lys (2.6%) and Met (0.9%), with no additional benefit from Met beyond 0.6% (Castro et al., 2016). Spray-dried plasma proteins (**PP**), enriched with immune proteins and growth factors (Campbell et al., 2008; Pérez-Bosque et al., 2016), supply most EAA except Lys, Met, and Ile while retaining albumin and IgG functionality; replacing up to 33% of milk proteins with PP sustained calf performance (Vasquez et al., 2017). Morrison et al. (2017) found in a 56-d trial with calves fed MR (22% CP, 20% fat, 2.0% Lys) at 1.5%–1.75% BW (DM basis) that 10% PP without Ile and Thr supplementation reduced ADG (0.54 vs. 0.60 kg/d), gain:feed ratio, and final BW (78.3 vs. 81.9 kg). However, supplementing Ile and Thr maintained BW, improved feed efficiency, and decreased both diarrhea incidence and antibiotic use compared with the all-milk controls. These findings underscore the importance of optimizing AA composition in alternative protein sources to ensure efficient utilization and maintain growth performance in preweaning calves.

Amino acid supplementation increases the resilience of calves to immunological challenges beyond supporting growth (Zubia et al., 2023), as inflammation increases the need for AA for the synthesis of acute-phase proteins, antibodies, and glucose precursors, thus diverting resources from growth (Li et al., 2007). In a 45-d trial with Holstein bull calves (28 d old, 44 kg BW) fed MR (20% CP, 20% fat DM basis), Zubia et al. (2023) found that supplementation with 10 EAA after an LPS challenge increased plasma Met (+25%), Leu (+18%), Thr (+22%), and anti-ovalbumin IgG titers (+15%) while reducing glucose declines (−6.8 mg/dL less) and cortisol spikes compared with nonsupplemented controls. These results underscore the importance of a precise AA supplementation strategy that considers developmental stage, health status, and protein source to optimize growth, N efficiency, and immune system development. Ongoing research is essential to refine AA requirements and alternative protein formulations for sustainable calf rearing.

OPTIMIZING PROTEIN NUTRITION IN CALF STARTERS: THE ROLE OF CP LEVELS AND FEED PROCESSING

One of the most discussed aspects of calf nutrition is the optimal CP content in the starter diets. Numerous studies have investigated the effects of different protein levels on growth performance, feed efficiency, and metabolism of calves, highlighting the complexity of CP requirements of young calves. Leibholz and Kang (1973) conducted a basic study in which male calves were fed a basal diet of 12% CP, with meat meal, soybean meal, or urea added to pelleted starters to increase CP levels to 15% or 18% (DM basis). The results showed that increasing the CP content from 12% to 18% significantly ($P < 0.05$) improved weight gain (mean range: 650–690 g/d), feed intake (mean range: 1.79–1.97 kg/d), and N retention, with the highest growth observed in calves fed the 18% CP diet. This early research established a clear link between higher protein levels and better growth performance in calves. Building on these results, Akayezu et al. (1994) investigated calf starter diets based on cracked corn (40%–52% of DM) without hay with a CP level between 15% and 22.4% DM basis and observed a positive correlation between greater CP contents and ADG (0.37, 0.39, 0.38, and 0.44 kg/d, respectively). After weaning, the 19.6% CP diet resulted in the highest gains (mean value: 0.86 kg/d), whereas diets above 22.4% showed no additional benefit, suggesting a protein threshold (Akayezu et al., 1994) under the conditions of this study. Hill et al. (2007a) carried out further studies on Holstein bull calves fed pelleted starter feed with a CP level of 15% to 26% (as-fed basis), finding that increasing the CP levels above 18% did not sustainably improve growth performance (ADG mean range: 668–688 g/d), feed efficiency, or body condition. Their study found that, although CP levels below 18% were suboptimal for growth (788 g/d), the benefits diminished with higher CP levels, particularly above 18% with well-texturized starters. A quadratic effect was observed between 15% and 21% CP, confirming that yields decrease after 18% CP.

Kazemi-Bonchenari et al. (2018) extended this study and investigated ground starter diets (without forage) with 18% and 20% CP, DM basis, in preweaning Holstein calves. Their study found no significant differences in overall starter feed intake (mean range: 712–734 g/d), overall ADG (mean range: 502–554 g/d), or final BW (mean range: 76–78 kg) on d 80 between the 2 CP levels, although rumen fermentation was slightly increased in the 20% CP (DM basis) diet, with higher molar proportions of butyrate and propionate and lower acetate in the rumen (Kazemi-Bonchenari et al., 2018). Luchini et al. (1991) assessed pelleted starter diets (20% vs. 25% CP, DM basis) in Holstein calves weaned at 26 or 42 d, with MR offered at 10% of birth weight, finding no effect of CP level on feed intake or scour incidence. Adding to this body of research, Boorboor et al. (2020) examined ground starter diets (without forage) with 16%, 18%, and 20% CP DM basis. Although no significant differences in growth or feed efficiency were observed, calves fed the 16% CP diet exhibited reduced plasma albumin and urea N levels, suggesting potential metabolic inefficiencies (Boorboor et al., 2020).

Other studies provide valuable insights into optimizing protein content in calf starter diets, showing that although protein is a critical factor, its effectiveness is often influenced by other components such as energy intake and milk allowances. Stamey et al. (2012) investigated the effects of CP content in well-textured starter diets (19.6% vs. 25.5% CP DM basis) on Holstein calves fed either a high-protein MR (28.5% CP, 15% fat, DM basis) or a conventional MR (20% CP, 20% fat, DM basis). The conventional MR was fed daily at 1.25% of birth weight in 2 feedings in wk 1 to 5 and once daily at 0.625% of birth weight in wk 6. In contrast, the high-protein MR was fed 1.5% of birth weight as DM in wk 1 and 2 and 2% of birth weight as DM in wk 2 to 5, divided into 2 daily feedings (Stamey et al., 2012). Calves fed improved MR with high-CP starter diets gained more weight and had greater BW at wk 8 and 10. They found that a 27% reduction in CP decreased ADG and FE, and the optimal CP-to-energy ratio to maximize ADG was 51.5 g CP/Mcal ME at 3.26 Mcal/d and 55.0 g CP/Mcal ME at 3.71 Mcal/d. However, there was a wide CP percentage range between the 2 starters, with no intermediate CP levels. Building on this, Stamey Lanier et al. (2021) investigated the effects of CP content in pelleted starter diets on the growth and body composition of Holstein calves fed either a low feeding rate of MR (20.6% CP, 21.7% fat, DM basis) with a low-CP starter diet (21.5% CP), a high feeding rate of MR (29.1% CP, 17.3% fat, DM basis) with a low-CP starter diet, or a high feeding rate of MR with a high-CP starter diet (26% CP, DM basis). Calves fed high-rate MR gained more weight and had higher BW at wk 8 and 10 with greater carcass weights at wk 5 and 10. High-CP starters increased plasma BHB, indicating improved gastrointestinal tract development, whereas elevated plasma urea N levels indicated a suboptimal AA profile in the starter or insufficient energy intake relative to protein supply, or both (Stamey Lanier et al., 2021). In addition, high-protein starters optimized the composition of EBW gain and increased reticulorumen weight at 10 wk of age (Stamey Lanier et al., 2021). However, there was a wide CP percentage range between the 2 starters, with no intermediate CP levels. Kazemi-Bonchenari et al. (2022) investigated the effects of moderate versus high levels of milk feeding and a finely ground starter containing 18% or 23% CP (DM basis, with 7% finely chopped wheat straw) on growth performance and urinary purine derivatives in Holstein calves and found that a ground starter containing 23% CP improved feed intake, ADG, and microbial protein synthesis before weaning, but decreased N utilization after weaning. They showed that feeding a high amount of milk together with a ground starter with a higher CP content increased withers and hip heights, elevated blood insulin concentration, and optimized the CP-to-ME ratio. Despite the large difference in CP levels between the 2 starters and the lack of intermediate CP levels, the results suggest that greater CP content in ground starters may improve growth and nutrient utilization in calves with a high level of milk feeding (Kazemi-Bonchenari et al., 2022). Yazdanyar et al. (2025) studied Holstein calves (40.6 kg BW) fed high or moderate levels of milk with ground starter (7.5% chopped wheat straw) varying in CP (20% vs. 24% DM basis) and fat (3.0% vs. 5.5% DM basis) until 73 d. High milk with 24% CP yielded the highest preweaning (815 g/d) and overall (867 g/d) ADG, increased ruminal propionate and butyrate, and increased urinary purine derivatives, indicating greater microbial protein synthesis. However, 5.5% fat reduced preweaning starter intake (380 g/d) and microbial efficiency, although 24% CP (DM basis) mitigated N deficits. These results indicate that an optimal CP level in calf starters, generally between 20% and 23% CP

(DM basis), with moderate milk feeding and increasing to 24% to 25% (DM basis) with greater milk feeding, significantly promotes calf growth and development of early-stage calves.

The physical form and processing of starter diets significantly influence their efficacy in promoting growth and rumen development (Ghaffari and Kertz, 2021; Ghaffari et al., 2025). Makizadeh et al. (2020) investigated the influence of protein content and feed processing on growth and skeletal development. Although increasing CP from 18% to 21% (DM basis) did not significantly affect ADG (mean range: 634–714 g/d), starter feed intake (mean range: 882–959 g/d), or FE, combining the 21% CP (DM basis) diet with steam-flaked corn in starter resulted in improved skeletal growth, as evidenced by increased withers and hip heights (Makizadeh et al., 2020). Sahib et al. (2023) investigated the effects of 55% steam-flaked corn and 55% steam-flaked barley with 18% and 22% CP levels (DM basis) in meal calf starters (including 5% finely chopped wheat straw) on growth, microbial protein synthesis, and inflammatory markers in Holstein calves. Sahib et al. (2023) found that the 22% CP diet increased preweaning ADG by 144 g/d, starter feed intake by 104 g/d, and microbial protein synthesis, especially when paired with steam-flaked corn, whereas the 18% CP (DM basis) diet with steam-flaked barley resulted in decreased microbial protein synthesis and increased inflammatory markers.

Overall, optimizing protein nutrition in calf starter diets requires a balance between protein and energy intake due to their interdependence, as increasing the CP content, which is usually above 20% to 22%, often reduces ME density by replacing energy-rich components such as cereal grains and fats with protein-rich components such as soybean meal or other proteins. This energy dilution, exacerbated by excess N from soybean meal metabolism that increases hepatic ureagenesis and urinary N excretion costs, reduces ME available for growth, negating the benefits of greater CP content in calf diets. In addition, the suboptimal AA profile of soybean meal, which is low in Met, Lys, and Thr, restricts protein synthesis and exacerbates N inefficiency. Therefore, relying solely on CP or even the CP/ME ratio is not sufficient, as these metrics do not capture AA availability and balance, and overlook the critical interactions between protein intake, energy utilization, and growth efficiency.

METABOLIZABLE PROTEIN DYNAMICS IN CALVES: RUP AND RDP CONTRIBUTIONS

Metabolizable protein represents the protein fraction available for absorption in the small intestine and provides AA necessary for maintenance, growth, and production (Burroughs et al., 1974; NASEM, 2021). Metabolizable protein includes RDP and RUP, where RDP is hydrolyzed by ruminal microbes to ammonia and peptides to support microbial crude protein (**MCP**) synthesis, which is optimized by synchronizing RDP with fermentable carbohydrates and is influenced by protein solubility and degradability, and AA composition (Russell et al., 1992; NASEM, 2021). As reported by NASEM (2021), the relative contributions of RDP and RUP to MP shift substantially between 30 and 600 d of age, corresponding to changes in BW (65 to 530 kg), growth rate, and rumen development. Although RDP remains constant at 10% of the ration DM, CP concentration decreases from 21.0% at 30 d of age to 12.7% at 600 d of age. Consequently, the proportion of RDP in CP increases from 47.6% to 78.7% (100 to 600 d of age), reflecting the increasing reliance on MCP as calves age. In contrast, the proportion of RUP decreases from 6.6% to 2.7% of the ration DM and its proportion in CP from 31.43% to 21.26% (100 to 600 d of age), indicating a lower dietary requirement for bypass protein in the diets of older calves.

In preweaning calves, MP supply primarily originates from milk proteins, as MCP production remains minimal due to limited ruminal fermentation. A meta-analysis of calves aged 4 to 76 d (Quigley et al., 2021) reported that apparent total-tract N digestibility increased from 83% at birth to 92.7% after 29 d (n = 47 studies, 312 observations) and then stabilized due to increased enzyme activity (e.g., doubled chymotrypsin after 3 wk) and intestinal maturation (Guilloteau et al., 2009). Because liquid feed (milk or MR) bypasses the rumen, the microbial N contribution remains negligible during this phase (Quigley et al., 1985; NASEM, 2021). NASEM (2021) provides standardized conversion factors for dietary CP to MP, accounting for calf diet, rumen function, and postruminal digestion. These factors are 0.95 for milk or milk-derived ingredients, 0.75 for combined milk and starter diets, and 0.70 for fully ruminated calves on starter diets only. For milk-fed calves with access to starter, the CP-to-MP conversion is a weighted average of the dietary components. The value of 0.95 represents the true digestibility of milk proteins in young calves (NASEM, 2021); however, the 0.95 factor may overestimate protein digestibility in calves under 2 to 3 wk old, due to their immature digestive enzyme system (Terosky et al., 1997). In addition, the protein source influences endogenous losses and digestibility. Plant proteins in MR have been associated with increased endogenous protein losses that reduce apparent ileal digestibility compared with skim milk powder-based MR (Montagne et al., 2000, 2001; Ansia and Drackley, 2020). Therefore, these increased endogenous losses must be taken into account when calculating the MP for feeds containing plant proteins (Montagne et al., 2001; NASEM, 2021).

The introduction of starter feed enhances MCP synthesis by supplying fermentable carbohydrates, thereby increasing microbial N contribution (Leibholz, 1975; Lallès and Poncet, 1990). As rumen function develops, MCP synthesis increases and reaches a plateau at ~60.6% (SE = 1.43) of total N intake once DMI of calves exceeds 1.32 kg/d (NASEM, 2021). This threshold means that the rumen is increasingly able to maintain protein supply via microbial sources as calves transition from a milk-based diet to solid feed postweaning (NASEM, 2021). Microbial composition also influences MCP dynamics. In mature cows, protozoa make up 40% of the microbial biomass in the rumen (Russell and Rychlik, 2001). Recent data suggest that protozoa contribute on average 23% of the microbial flux of EAA, with contributions ranging from 17% for Trp to 28% for Lys (Dineen et al., 2021). Protozoa exhibit a distinct AA profile that includes a higher proportion of Ile, Lys, Asp, Glu, Cys, Phe, and Tyr, but less Thr, Arg, Gly, Ala, Val, and Ser, based on the rumen contents of cattle (Martin et al., 1996). In ruminants, colonization with protozoa begins around 15 d of age and requires transmission through saliva, which may be delayed in isolated or individually housed animals (Fonty et al., 1988; Chaucheyras-Durand et al., 2019; Morgavi et al., 2020). These developmental and microbial shifts influence the AA profile of MCP and emphasize the need for precise MP estimation across different growth stages.

During the weaning period, calves undergo a pivotal nutritional shift as their reliance transitions entirely from liquid feeds (e.g., milk or MR) to solid feeds, with rumen MCP emerging as the primary source of MP and AA. During the weaning process, milk intake declines, and DMI is often insufficient to support lean tissue growth, increasing dependence on rumen fermentation to meet AA requirements (Van Amburgh et al., 2019). After weaning, rumen development accelerates dramatically. For example, in Holstein bull calves weaned at 5 wk of age, rumen volume more than doubled (from 5.8 to 13.2 L), BW increased by 42%, and N utilization improved substantially (Vázquez-Añón et al., 1993). This rapid morphological and functional maturation underlines the increasing contribution of MCP to the calf's AA supply during the transition to a solid diet.

The AA profile of MCP evolves during the postweaning period and is influenced by changes in diet and rumen microbial dynamics. Quigley et al. (1985) demonstrated this transition in Holstein bull calves fed pelleted starter feeds and weaned at 4 or 8 wk, showing that by 5 to 7 wk, bacterial N from rumen microbes accounted for 40% to 50% of abomasal N, indicating rapid rumen maturation and replacement of RUP from liquid feed by MCP as the primary source of AA. In their study, Met was consistently limiting in abomasal digesta (5.2–5.4 g/100 g EAA), Lys decreased from 16.9 to between 13.7 and 14.6 g/100 g, and Arg increased from 7.1 to between 10.1 and 10.9 g/100 g (all values expressed on a DM basis) as starter feed intake increased and milk intake decreased, reflecting starter-induced changes in MCP composition (Quigley et al., 1985). Further evidence comes from 2 extensive studies (Abe et al., 1998, 2000) on limiting AA in Holstein calves fed restricted corn and soybean meal diets. In the first study (Abe et al., 1998), calves over 3 mo of age were fed a basal diet of 20 or 27 g/kg BW (DM basis), resulting in ADG of 0.60 to 1.31 kg/d. The addition of 0.111 g of DL-Met and 0.333 g of L-Lys HCl per kilogram of BW (DM basis) increased the Met content in plasma from 2.8 to 20.7 mmol/dL; N retention increased from 44% to 48%, N losses in urine decreased, and branched-chain AA in plasma were reduced. In the second study (Abe et al., 2000) with weaned calves under 3 mo of age, 4 trials showed that replacing nonspecific N with DL-Met and L-Lys HCl increased N retention (e.g., from 40.8% to 46.9% in one trial); and optimizing DL-Met to 0.11 g/kg BW versus 0.022 g/kg BW (DM basis) and adding L-Trp (0.055 g/kg BW DM basis) further increased retention to 54.3%. These results suggested Met as the first-limiting AA, Lys as probably second limiting, and Trp as potentially a co- or third-limiting nutrient, emphasizing the importance of precise AA supplementation to improve N efficiency and growth in calves (Abe et al., 2000). Rumen-protected AA studies refine these findings: rumen-protected Met and the analog HMBi (isopropyl ester of 2-hydroxy-4-(methylthio)-butyrate acid) increased plasma Met and starter intake (+15%) in Holstein calves (14–91 d) fed a pelleted starter (25% CP and 2.5 Mcal of ME/kg of DM basis), although Met was not limiting in the basal diet, whereas HMTBa (2-hydroxy-4-(methylthio)butanoate) depressed intake and growth without elevating plasma Met levels (Molano et al., 2020). Despite these insights, optimal AA in calf starter diets remain poorly characterized.

The RDP content in the diet varies considerably between feedstuffs and influences MCP and N utilization in calves. Cereal grains such as barley grain, corn, and oats contain lower levels of RDP than forages such as early-season alfalfa hay, timothy hay, and corn silage (NASEM, 2016), but the limited inclusion of forages (5%–15% of diet DM; Imani et al., 2017) in the preweaning diet limits their RDP contribution. Therefore, protein sources such as soybean meal and canola meal are the main sources of RDP in calf starter feeds. The solubility and structural properties of proteins regulate the availability of RDP, which is categorized into an immediately soluble fraction, a potentially degradable fraction, and a nondegradable fraction (Van Soest et al., 1981). Peptide structure further modulates the rate of degradation, as Lys-Pro degrades 5 times slower than Lys-Ala, and Pro-Met degrades 2.5 times slower than Met-Ala (Yang and

Russell, 1992). This is particularly important in calves, as their developing rumen microbiota and enzyme system benefit from a gradual and sustained release of degradable proteins to optimize microbial growth and MCP synthesis while minimizing ammonia accumulation due to rapid degradation. In addition, heat treatment (as with heat-treated soybean meal; Kazemi-Bonchenari et al., 2016) decreases the solubility of proteins through Maillard reactions, thereby increasing the RUP and decreasing the bioavailability of Lys (Van Soest, 1994) These properties must be taken into account in the formulation of calf starter diets to balance microbial requirements with the postruminal availability of AA.

Synchronizing RDP with fermentable carbohydrates improves MCP synthesis by optimizing N and energy availability, reducing ruminal ammonia losses, and improving rumen efficiency (NASEM, 2021). Abdelgadir et al. (1996a) found that in Holstein calves (0.5–8 wk), different levels of RUP in iso-nitrogenous starters (soybean meal vs. roasted soybeans; raw corn vs. gelatinized conglomerated corn; ± 1% urea) had no effect on performance, although urea decreased growth in conglomerated corn and soybean meal diets while increasing rumen ammonia and plasma urea, depending on starch and protein sources. Synchronous versus asynchronous CP-starch pairings resulted in similar growth outcomes, with shifts in plasma AA (e.g., doubled citrulline in soybean meal-conglomerated corn; increased Phe in roasted soybean), suggesting that the benefits of synchrony are limited at this age. In another study, Abdelgadir et al. (1996b) demonstrated that in calves fed pelleted 18% CP starters, as-fed basis (soybean meal [30% RUP], roasted soybeans at 138°C [45% RUP], or 146°C [52% RUP] with raw or roasted corn), synchronization of ruminal availability (e.g., roasted soybeans at 146°C with raw corn) improved BW gain and feed conversion, but asynchronous diets (e.g., roasted soybeans at 146°C with roasted corn) were less effective. Maiga et al. (1994) observed that Holstein calves (1–12 wk) fed pelleted diets containing 18% CP (DM basis) diets, with 20% ground alfalfa hay, had modest gains in BW when fed less degradable carbohydrate and more RUP (e.g., corn with extruded soybean meal), likely due to higher DMI, suggesting improved palatability and rumen efficiency with synchronized nutrients. Chishti et al. (2021) extended these findings in 8- to 16-wk-old calves by showing that growth and digestibility were influenced by starch as a function of dietary CP content. Calves fed a high-starch, high-protein diet (38% starch and 22% CP, DM basis) had higher total-tract digestibility in terms of DM (+11%), NDF (+9%), and ADF (+11%) and higher ruminal pH (+0.51 units 20 h after feeding) and greater in situ disappearance of wheat middlings (DM: 71.2% vs. 55.2%) and soybean meal (DM: 74.5% vs. 62.5%). In contrast, high-starch, low-protein diets (38% starch and 16% CP, DM basis) decreased the in situ NDF digestibility of wheat middlings by up to 44% and protein digestibility of soybean meal by 14% after 24 h, and decreased the DM digestibility of grass hay (−8%; Chishti et al., 2021). These diets also increased the total concentration of VFA in the rumen by 20% 14 h after feeding, elevated propionate molar proportion by 18%, and decreased the acetate:propionate ratio by 26%, indicating improved fermentation efficiency but suppressed cellulolytic activity (Chishti et al., 2021). Overall, these studies suggest that synchronization of RDP and carbohydrate fermentability promotes MCP synthesis and growth, and that the effects of RUP and carbohydrate degradability are context-dependent and need to be adjusted to the rumen developmental stage for optimal nutrient utilization.

Beyond synchronization, an appropriate RUP-to-RDP balance enhances N retention by providing bypass protein for direct absorption, particularly when microbial protein synthesis is limited. Kazemi-Bonchenari et al. (2020) reported that increasing the RUP:RDP ratio from 27:73 to 38:62 in ground starter feeds with a constant CP content of 20% (DM basis) improved overall ADG by 103 to 173 g/d and final BW by 7.2 to 7.7 kg/d in Holstein calves. Rastgoo et al. (2020) reported that starter feed intake (mean range: 925–998 g/d), ADG (mean range: 652–751 g/d), and BW were not significantly influenced by the RUP:RDP ratio (low ratio = 28:72; high ratio = 37:63) in ground starter diets containing 20% CP (DM basis) and 5% finely chopped wheat straw, but feeding steam-flaked corn (57% of DM) in a ground starter with the higher RUP:RDP ratio (37:63) significantly improved skeletal growth. Notably, both withers height and hip height increased in the group with high RUP:RDP ratio compared with the low-ratio group, suggesting that higher RUP intake promotes skeletal development in growing calves under the conditions of this study. The study by Yousefinejad et al. (2021) examined the effects of 2 CP levels (18% vs. 22%; DM basis) and varying RUP:RDP ratios (low ratio, 26:74, vs. high ratio, 35:65) in finely ground starter diets on Holstein calves. The results showed that a 22% CP diet increased N intake, RUP intake, ADG, weaning BW, FE, and microbial protein synthesis compared with the 18% CP diet. Although the RUP:RDP ratio did not significantly affect feed intake or preweaning growth, calves fed higher RUP levels (35:65) showed improved postweaning ADG (by 50–52 g/d), further emphasizing the importance of RUP during critical growth phases. In this study, the results may have been influenced not only by the RUP:RDP ratio but also by the higher fermentability of the ground ingredients and the increased CP deamination, which could affect ruminal ammonia concentration and pH levels. Additionally, increasing CP content to 22% improved ruminal

Table 2. Energy and protein requirements for Holstein heifers with mature BW of 700 kg based on NASEM (2021) equations

Item	Live BW (kg)				
	112	224	336	420	560
BW, as percentage of mature BW	16	32	48	60	80
Estimated DMI (kg/d)	3.3	6.0	8.0	9.3	10.9
Maintenance ME (Mcal/d)	5.5	9.3	12.6	14.8	18.4
Maintenance MP (g/d)	119	225	322	388	484
ME for gain (Mcal/d; ADG: 700–980 g/d)	8.8–10.2	13.3–14.9	17.3–19.3	20.2–22.3	28.8–31.3
MP for gain (g/d; ADG: 700–980 g/d)	433–498	599–672	711–790	767–846	952–1,034
Minimum MP/ME	49	45	41	38	33
ME/kg diet DM (ADG: 700–980 g/d)	2.6–2.3	2.2–2.5	2.2–2.4	2.2–2.4	2.6–2.9
CP, percentage of diet DM (ADG: 700–980 g/d)	18.4–21.1	14.3–16.0	12.6–14.0	11.8–13.0	12.5–13.5

fermentation, particularly enhancing short-chain fatty acids and the acetate-to-propionate ratio in the rumen, which are indicators of improved rumen function and microbial activity (Yousefinejad et al., 2021). Hill et al. (2006) studied Holstein calves (42 ± 2 kg) fed 18% CP as-fed starters (37% corn, 35% supplement pellets, 25% oats, 3% molasses) with a 30% increase in RUP and a 10% increase in MP and found no significant differences in ADG (0.55 ± 0.04 kg), starter feed intake (2.31 ± 0.14 kg), or gain-to-feed ratio over 56 d, suggesting that these theoretical values for the feeds are not suitable for young calves or that protein fractions were not a limiting factor for growth. Overall, the dynamics of MP in calves are driven by the interplay between RDP and RUP, with efficient MCP synthesis playing a central role in the availability of N and AA, especially as rumen function matures during the postweaning period.

PROTEIN REQUIREMENTS FOR REPLACEMENT DAIRY HEIFERS

Replacement dairy heifers undergo pronounced physiological and metabolic changes between weaning and first calving that require stage-specific refinement of dietary protein and energy supply to support lean tissue accretion, skeletal development, and future lactation capacity (Van Amburgh et al., 2019). As heifers increase from 16% to 80% of mature BW (112 to 560 kg, based on a mature BW of 700 kg for Holsteins), their DMI increases from 3.3 to 10.9 kg/d in response to greater rumen capacity and increased metabolic demand (NASEM, 2021; Table 2). Concurrently, maintenance ME and MP requirements increase from 5.5 to 18.4 Mcal/d and from 119 to 484 g/d, respectively (NASEM, 2021; Table 2), reflecting the proportional increase in basal metabolic load. Nutrient requirements for growth are maximal during intermediate stages of development, with ME requirements for gain increasing from a range of 8.8 to 10.2, to between 28.8 and 31.3 Mcal/d, and MP requirements for gain increasing from a range of 433 to 498, to 952 to 1,034 g/d, to support tissue synthesis at an ADG of 700 to 980 g/d (NASEM, 2021; Table 2). As dairy heifers approach maturity, they physiological transition from predominantly protein-driven muscle accretion to energy-driven fat deposition. This is reflected in a progressive decrease in the MP-to-ME ratio (from 49 to 33), a transient stabilization of dietary ME density (2.2–2.4 Mcal/kg DM), followed by an increase to a range of 2.6 to 2.9 Mcal/kg DM, and a decrease in dietary CP from a range of 18.4% to 21.1%, to between 12.5% and 13.5% (NASEM, 2021; Table 2). These shifts primarily reflect improved efficiency of N utilization through improved MCP synthesis rather than a reduction in AA requirements per se. This underscores the importance of synchronizing fermentable energy with the degradable and undegradable protein fractions to optimize nutrient intake and support heifer development.

For prepubertal heifers, optimizing the CP-to-ME ratio is crucial. Research by Lammers and Heinrichs (2000) investigated the effects of different CP-to-ME ratios (46:1, 54:1, and 61:1 g/Mcal) on the growth of heifers from 28 to 48 wk of age. They found that increasing this ratio from 46:1 to 61:1 g/Mcal improved feed efficiency by 6% and ADG by 9%. However, exceeding this threshold provided no additional benefits, indicating a practical limit to protein supplementation (Lammers and Heinrichs, 2000). Similarly, Gabler and Heinrichs (2003) studied prepubertal Holstein heifers (125–234 kg BW) on a restricted diet targeting an ADG of 0.8 kg/d and found that a diet with a CP:ME ratio of 59 to 68 g/Mcal optimized skeletal growth and protein utilization, whereas lower ratios (48 g/Mcal) restricted growth, and higher ratios (77 g/Mcal) resulted in inefficiencies that increased ruminal ammonia by 30% and urinary N excretion by 25%. Whitlock et al. (2002) evaluated the effects of dietary protein intake on growth and mammary development in Holstein heifers fed ad libitum high-energy diets from ~3.5 to 8.9 mo of age (~134 to 320 kg BW). Diets contained 48, 57, or 66 g CP/Mcal ME (equivalent to 37, 41, or 44 g MP/Mcal ME), with protein density increased by incorporating expeller soybean meal. This approach elevated the RUP:RDP ratio and shifted the source of ME toward AA oxidation and away from carbohydrate-derived energy (Whitlock et al., 2002). Although overall growth performance, carcass composition, and endocrine markers (somatotropin and IGF-I) were unaffected, mammary parenchymal development was compromised in heifers fed the low-protein diet, particularly those that reached puberty early (Whitlock et al., 2002). Chelikani et al. (2003) investigated the effects of dietary protein density on reproductive maturation in Holstein heifers by comparing high- (20.9% CP), medium- (18.1% CP), and low-CP (13.5% CP) diets. Heifers fed the high-CP diet reached puberty earliest (9 vs. 11 vs. 16 mo for high-, medium-, and low-CP diets, respectively) despite similar body composition at puberty. High-CP heifers exhibited greater backfat thickness, increased LH pulse frequency at 10 mo, and larger dominant follicle diameters (12.8 vs. 12.2 vs. 10.6 mm).

The balance between RDP and RUP plays a critical role in N utilization and growth efficiency in heifers, with optimal proportions adjusting as rumen function develops. Zanton et al. (2007) assessed postpubertal heifers (455 ± 4.0 kg BW) fed diets with varying soluble protein (26.6%–37.7% of CP) and RUP levels (38.1%–46.7% of CP), finding that elevating soluble protein increased rumen ammonia by 4.5 mg/dL, whereas raising RUP improved N retention by up to 12.3%. However, diets with high soluble protein and high RUP led to the highest total N excretion, indicating that excessive protein degradability can compromise N efficiency in postpubertal heifers (Zanton et al., 2007). Silva et al. (2018a,b) investigated the effects of increasing levels of RUP (38%, 44%, 51%, and 57% of CP in the diet; DM basis) on prepubertal (106 ± 7.6 kg BW, 4.3 ± 0.46 mo) and pubertal (224 ± 7.9 kg BW, 12.6 ± 0.45 mo) Holstein heifers over a period of 84 d. No interaction between physiological stage and RUP level was observed (Silva et al., 2018a). Those authors found that increasing RUP from 38% to 57% of total dietary CP in Holstein heifers did not affect DMI, CP intake, or rumen digestion rates, but 51% and 57% RUP tended to increase MP flow and N retention while reducing urinary N excretion, with 51% RUP recommended for optimal growth (1 kg/d ADG; Silva et al., 2018b). Despite a decline in microbial protein synthesis and efficiency with higher RUP, the 51% RUP level balanced MP supply and N use efficiency without compromising intake or digestibility, supporting its use in growing heifer diets (Silva et al., 2018b). Further study by Corea et al. (2020) showed that increasing RUP from 26% to 36% of CP in heifers (166 kg BW) over 10 wk increased ADG (0.982 vs. 0.914 kg/d), feed efficiency (+6.3%), and N efficiency (+4.8%) but reduced MCP by 9.4%, indicating a tradeoff between RUP and microbial protein contribution.

Adjusting the ratio of RDP to RUP also affects N dynamics and long-term productivity. Tomlinson et al. (1997) demonstrated that increasing the proportion of RUP from 31% to 55% of total N in isonitrogenous, isocaloric diets (formulated for 0.60 kg/d ADG) for Holstein heifers (213–231 kg BW) over 50 d increased DMI (97.6 to 73.5 g/kg $BW^{0.75}$) and digestible energy intake (0.28 to 0.21 Mcal/kg $BW^{0.75}$ per day), while ADG (0.84 to 0.96 kg/d) and feed efficiency (20.6 to 13.3 Mcal DE/kg BW gain) were improved, with no change in body composition. The study suggests that a higher RUP content (e.g., 55%, administered via blood meal) improves AA supply to the lower intestine and thus increases growth performance. Lascano et al. (2016) investigated the effects of feeding a high-RDP diet (64%–71% of CP) with different forage-to-concentrate ratios (45% vs. 90% forage) and fiber proportions when feeding Holstein heifers (487 ± 15 kg BW). They found that N digestibility remained unchanged, whereas increasing fiber levels decreased N retention by up to 8.3% and decreased predicted MCP flux into the duodenum by 9.7%, highlighting the effects of fiber on protein utilization efficiency in growing heifers (Lascano et al., 2016). Capuco et al. (2004) found that the increase in RUP from 5.9% (14.9% CP) to 7.9% (16.9% CP, DM basis) in prepubertal Holstein heifers (90 d to 10 mo) along with administration of bovine somatotropin (**bST**; 0.1 mg/kg BW per day) induced rapid body growth (parenchymal mass increased from 16 to 364 g, DNA from 58 to 1,022 mg) without affecting mammary epithelial cell proliferation, parenchymal DNA, or lipid content, suggesting no adverse effects on mammary gland development. Subsequent milk production and composition remained unaffected by RUP or bST, suggesting that a modest increase in CP as RUP (16.9% vs. 14.9%) can improve frame size and growth efficiency without affecting future lactation performance (Capuco et al., 2004). Oney et al. (2016) investigated the effects of increasing RUP levels (0%, 3.25%, 6.5%, 9.75%, and 13%) on 60 steers aged ~6 mo, with an initial weight of 290 kg, that were fed silage for 84 d. The study found that higher RUP levels led to linear improvements in ADG and gain-to-feed ratio, with the highest RUP level (13%) leading to significantly greater ADG compared with lower levels (Oney et al., 2016). Final weight also increased with higher RUP addition, reaching 503 kg in the 13% RUP group, whereas the 0% RUP group reached 478 kg, with no differences observed in DMI (Oney et al., 2016). These findings underscore the importance of adjusting RUP and RDP levels based on the developmental stages of the calf, ensuring that protein sources are optimized to support growth and efficiency.

CONCLUSIONS AND FUTURE RESEARCH DIRECTIONS

In summary, optimizing protein intake in the diet requires a balance between protein and energy intake to support lean tissue growth while minimizing N excretion and excessive fat deposition. Recent studies emphasize the importance of CP-to-ME ratio, protein quality, and protein sources in MR and starter feeds, especially in the period prior to and around weaning. As calves shift from milk to solid diets after weaning, the AA profile of the MCP changes. This requires precise supplementation of key AA (e.g., Met, Lys, Trp) and synchronization of RDP with fermentable carbohydrates to increase MCP synthesis and N efficiency. In addition, optimizing the ratio of RUP to RDP promotes growth through the supply of bypass protein, although the benefits vary depending on the stage of development.

However, significant gaps still exist in our understanding of protein dynamics at critical developmental stages, particularly in the period from postweaning to first calving. Many studies focus solely on feeding milk or MR, but CP degradation of starter feed and AA availability vary widely due to differences in DMI and protein sources, making quantitative assessment difficult, as starter feed intake typically increases as calves ages. Future research should focus on determining the optimal ratio of RDP to RUP for microbial protein synthesis in the rumen and evaluating the supplementation of AA in the initial diet in the period from postweaning to first calving. Studies on the changes in body composition in the pre- and postweaning phases are needed to clarify how protein intake affects lean tissue growth, adiposity, and skeletal development. Linking early-life protein strategies to postweaning growth, reproductive maturity, and first-lactation performance remains an important research gap that requires long-term studies to track protein requirements, body composition, and utilization.

In addition, the physical form of the starter diet (e.g., pelleted, textured, ground) has important effects on DMI, ADG, rumen development, and rumination, indicating that CP percentage alone inadequately reflects starter feed quality (Porter et al., 2007; Ghaffari and Kertz, 2021; Kertz, 2025). Kertz (2023) posited that minimally processed grains, such as cracked barley, could provide sufficient fermentability, offering cost-effective alternatives to more extensively processed grains such as steam-flaked corn. Cracked barley can supply the necessary nutrients for microbial fermentation in the rumen, promoting growth without the added cost associated with steam-flaking. This emphasizes the importance of balancing both CP content and feed processing methods with overall physical form of starters to achieve optimal growth and cost-effectiveness. Furthermore, the inclusion of fiber in starter diets may influence protein requirements. Kertz et al. (1983) and Kertz (2010) found that high-fiber sources, such as wheat straw, might require higher protein levels to support effective fermentation and microbial activity in the rumen. Increased fiber content demands more protein to ensure sufficient microbial growth and N utilization. This interaction between fiber and protein content highlights the need to balance both components in starter diets to optimize rumen fermentation and microbial protein synthesis. Addressing these gaps will refine stage-specific nutritional strategies, ultimately improving the long-term productivity and reproductive performance of dairy heifers.

NOTES

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Nonstandard abbreviations used: bST = bovine somatotropin; EBG = empty body gain; EBW = empty BW; FE = feed efficiency; MCP = microbial crude protein; MR = milk replacer; PP = plasma proteins; SF = solid feed.

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